

RICARDO ALTAMIRANO MARQUEZ

AI Engineer | Agentic Systems & Full-Stack Infrastructure

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SUMMARY

AI Engineer and Full-Stack Engineer building production AI systems for domain-heavy operational workflows. Experienced delivering agent harnesses, tool-calling workflows, retrieval-augmented generation (RAG), structured outputs, multimodal analysis, and cloud-deployed applications using Python, FastAPI, React, SQL, Docker, AWS, and Azure.

SKILLS

Programming & Web: Python, TypeScript, JavaScript, Go, SQL, FastAPI, React, REST APIs, WebSockets

AI Systems: AI agents, agent harnesses, tool calling, RAG, structured outputs, multimodal analysis, multi-agent workflows, LangChain, LangGraph, vector databases

Cloud & Infra: PostgreSQL, MongoDB, Redis, Docker, AWS, Azure, GCP, Terraform, CI/CD, GitHub Actions

Platforms & Integrations: MCP-based tooling, Microsoft Copilot Studio, Vertex AI, Gemini Enterprise Agent, API integrations

EXPERIENCE

Hydrous Group | AI Engineer

Jan 2025 - Present | hydrousmgmt.com

- Built a vertical AI platform for water and waste operations that connected domain workflows with knowledge retrieval, tool execution, and structured outputs in production.
- Designed controlled agent workflows for water operations, enabling retrieval and deterministic calculations within governed business processes.
- Developed multimodal AI systems for water-treatment diagnostics using document, image, chart, and engineering-asset analysis with domain-grounded knowledge access.
- Built AI-powered technical proposal generation workflows with dynamic tables, charts, and process-flow diagrams for client-ready outputs.

DVAWEB | Software Engineer

Nov 2023 - Jan 2025

- Built and shipped end-to-end product features for an intelligent parking platform across architecture, backend services, frontend implementation, and production rollout.
- Developed AI-enabled internal tools, including CRM-focused knowledge retrieval and a natural-language-to-SQL system for non-technical teams.
- Built an enterprise knowledge platform over company documentation and prior projects, integrating proposal and CRM workflows in real time and reducing proposal turnaround time by 78%.

EDUCATION

Universidad Autonoma de Occidente | B.S. in Software Engineering